

Deep Learning-Based Prediction of Signal Peptides in Proteins Using Sequence and Physicochemical Feature Integration

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Abstract—Signal peptides (SPs) are short sequences found at the N-terminus of proteins that controls secretion of protein through the secretory pathway. Identifying signal peptides is critical for proteome annotation and identification of drug targets. Here, we introduced a deep learning based framework for identification of signal peptides using embeddings from a pre-trained BioBERT model. Here, we show that combining sequence embeddings with six physicochemical properties results in a better signal peptide prediction framework. Our model adds a 768-dimensional BioBERT protein sequence embedding with 6 physicochemical properties into a single joint embedding. This joint embedding is passed through a convolutional neural network (CNN) followed by a bidirectional LSTM. Binary prediction of signal peptides is then made from the next to the last layer. We trained the model on 1930 human protein sequences obtained from UniProt Swiss-Prot which were filtered before for redundancy to balance the number of signal peptides and non-Signal peptides. The model was tested using k-fold cross validation, with the k value being 5, with stratification and early stopping resulting in an average AUCROC of 0.8503 ± 0.0125 , MCC of 0.5383 ± 0.0460 , and F1-score of 0.7590 ± 0.0387 . These results show that joint embeddings of contextualized language models create a robust signal peptide classifier.

Index Terms—Signal Peptides, Deep Learning, CNN, BioBERT, BiLSTM, Protein Sequence Analysis

I. INTRODUCTION

Proteins are one of most fundamental workhorses of the cell, but for it to function correctly, it must be transported to the appropriate subcellular compartments. This targeting mechanism is frequently governed by a signal peptide—a short, N-terminal sequence of amino acids ranging from around 15 to 30 residues in length. Upon reaching the destination, the current sequences are typically cleaved from the mature protein. Identifying the presence of a signal peptide is a foundational step in genome annotation and recombinant protein production.

If we check historically, the experimental identification of signal peptides has been time-consuming and labor-intensive. Consequently, the computational prediction tools have become indispensable in molecular biology and also the traditional machine learning models relied heavily on manually engineered features based on the physicochemical properties of amino acids. However, the advent of Deep Learning (DL) and Natural Language Processing (NLP) has shifted the paradigm towards the learning representations directly from the raw sequences.

This paper majorly explores a multimodal feature integration approach. It extracts deep contextualized linguistic representations of protein sequences using BioBERT, a transformer-based language model, and concatenates them with standard physicochemical metrics. The integrated data is then passed through the Convolutional Neural Network coupled with a Bidirectional Long Short-Term Memory network (CNN-BiLSTM) equipped with an attention mechanism to classify the sequences as signal or non-signal peptides.

II. LITERATURE SURVEY

The transition from traditional machine learning to deep neural networks marked a significant milestone in the signal peptide prediction. Early deep learning models, such as DeepSig, demonstrated that the convolutional neural networks (CNNs) could substantially improve the detection of signal peptides and their cleavage sites compared to prior sequence-based methods [1]. This foundation was expanded upon by the SignalP 5.0, which played an important role in integrating deep neural networks to accurately classify the multiple signal peptide types across diverse biological domains [2]. The architectural evolution continued with frameworks like Signal-3L 3.0, which combined the attention mechanisms with Bi-

LSTMs and CNNs to capture complex sequence dependencies and improve window-based cleavage site scoring [3].

In recent years, the adoption of protein language models (PLMs) and transformer architectures has transformed the field by enabling predictions like organism-agnostic predictions. Models like SignalP 6.0 utilized large PLMs to predict all of the five types of signal peptides, successfully overcoming the limitations associated with species-specific biases and evolutionary divergence [4]. Similarly, USPNet demonstrates a high sensitivity and unbiased prediction capabilities across metagenomic datasets by leveraging deep PLM embeddings alongside the label distribution-aware margins [5]. The TSignal model further highlighted the efficacy of transformer attention mechanisms in refining both the overall classification and the precise localization of cleavage sites [6].

To address the computational resource demands inherent in deploying large PLMs, parameter-efficient fine-tuning (PEFT) strategies have been successfully applied for it. The PEFT-SP framework established that large PLMs could be adapted for signal peptide prediction with a minimal computational overhead, yielding significant performance gains even when it is trained on small or highly imbalanced datasets [7].

Beyond a pure sequence analysis, the explicit integration of physicochemical features has proven crucial for predicting secretion efficiency and providing biological interpretability. High-throughput experimental data coupled with machine learning has also enabled the quantitative evaluation of how specific the physicochemical properties and amino acid distributions influence protein export [?], [8]. These physicochemical insights are currently bridging the computational prediction and synthetic biology, with hybrid deep learning architectures being deployed for the *de novo* design of highly efficient and secretion-optimized signal peptides tailored for biotechnological applications [9].

III. MATERIALS AND METHODS

A. Dataset Preparation

The dataset was curated by retrieving sequences from the UniProt Knowledgebase (UniProtKB) with the help of REST API. In order to create a balanced dataset two queries were executed:

- Positive Class (Signal Peptides) Filtered using the query `ft_signal:* AND reviewed:true`.
- Negative Class (Non-Signal / Cytoplasmic) Filtered using the query `(cc_scl_term:Cytoplasm) AND (reviewed:true) AND (NOT ft_signal:*)`.

After retrieval of data, duplicate headers were removed to prevent data leakage. The final dataset consists of 1930 unique protein sequences where 980 of them are signal peptides and the remaining are 950 non-signal peptides. Therefore ensuring balance between both the classes

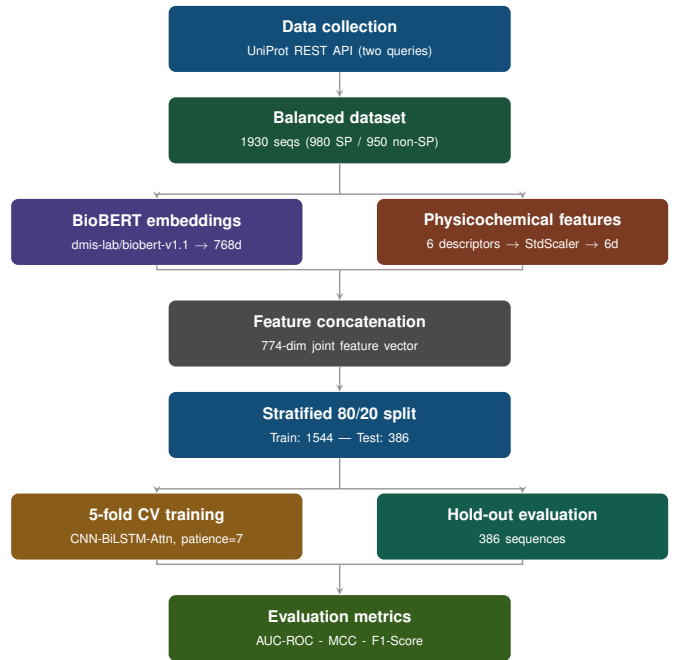
B. Feature Extraction

The proposed methodology has a two ways to extract features which captures both explicit biochemical metrics and latent sequence representations.

1) *Physicochemical Features*: The *Biopython* library was used to find out 6 different physicochemical properties the amino acid sequence of each protein: Grand Average of Hydropathy, Isoelectric Point, Molecular Weight, Instability Index, Aromaticity, and Secondary Structure Fraction (Helix). A standard scaler was used to make sure that a uniform variance is present within these values.

2) *Deep Sequence Embeddings*: The pre-trained `dmis-lab/biobert-v1.1` transformer model was used to convert these peptide sequences into vector embeddings. These sequences were first space-tokenized then reduced to a maximum length of 512 tokens and then it was passed through the model. A 768 dimensional vector for sequence was returned as the output of the final hidden state for the first token.

Finally this embedding vector of size 768 was combined with the physicochemical vector of size 6 thus giving a 774-dimensional robust feature vector that could be used to train the model ($X \in \mathbb{R}^{774}$).

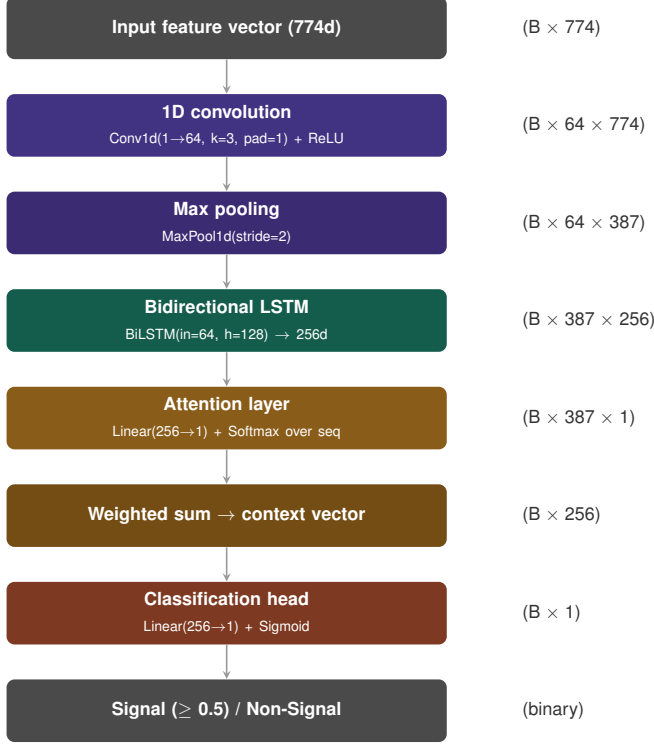


C. Model Architecture

A custom `CNNBiLSTMAttention` model was built using PyTorch to process the integrated features. The architecture is defined as follows:

- 1) **1D Convolutional Layer**: The 774-D input vector is unsqueezed into a 1-channel sequence. A 1D Convolution (64 filters, kernel size = 3, padding = 1) is applied, followed by a ReLU activation and a MaxPool1D layer (stride = 2).
- 2) **Bidirectional LSTM**: The pooled output is permuted and fed into a 1-layer BiLSTM with a hidden dimension of 128 (256 total due to bidirectionality).

- 3) **Attention Mechanism:** An attention layer computes softmax weights over the sequence length, creating a context vector that aggregates the most relevant temporal features.
- 4) **Classification Head:** The context vector is passed through a fully connected linear layer and a Sigmoid activation function to output a probability between 0 and 1.



IV. RESULTS & DISCUSSION

A. Model Implementation and Training

Model was trained using the Adam optimizer (learning rate $= 1 \times 10^{-3}$) and Binary Cross-Entropy loss. A batch size of 16 was utilized. Early stopping with a patience of 26 epochs was implemented to prevent overfitting, tracking the AUC-ROC on the validation set.

B. Results & Analysis

The model's generalization capabilities were evaluated using a Stratified k-Fold Cross-Validation strategy, where the value of k is 5. The performance metrics across all the folds is presented in the Table I.

TABLE I
5-FOLD CROSS-VALIDATION PERFORMANCE

Fold	AUC-ROC	MCC	F1-Score
Fold 1	0.8467	0.5532	0.7655
Fold 2	0.8619	0.5879	0.7872
Fold 3	0.8674	0.5819	0.8020
Fold 4	0.8408	0.4921	0.7500
Fold 5	0.8345	0.4763	0.6903
Mean $\pm$ Std	0.8503 $\pm$ 0.0125	0.5383 $\pm$ 0.0460	0.7590 $\pm$ 0.0387

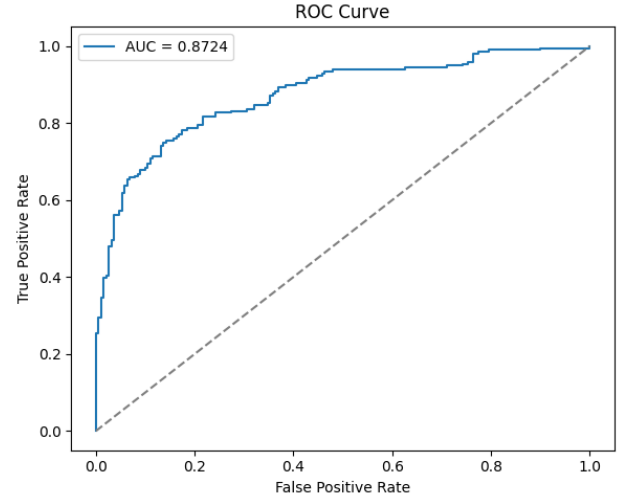


Fig. 1. ROC curve on the hold-out test set. AUC = 0.8724.

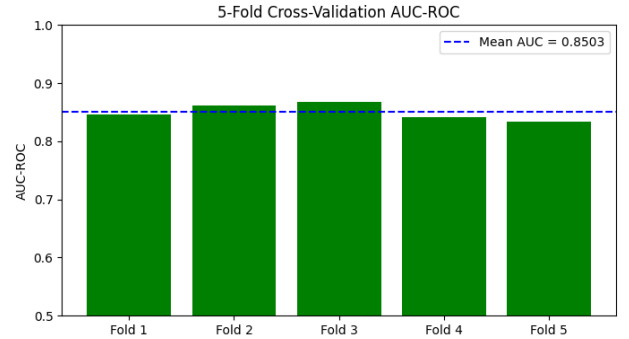


Fig. 2. Per-fold AUC-ROC scores across 5-fold stratified cross-validation. The dashed blue line shows the mean AUC of value 0.8503.

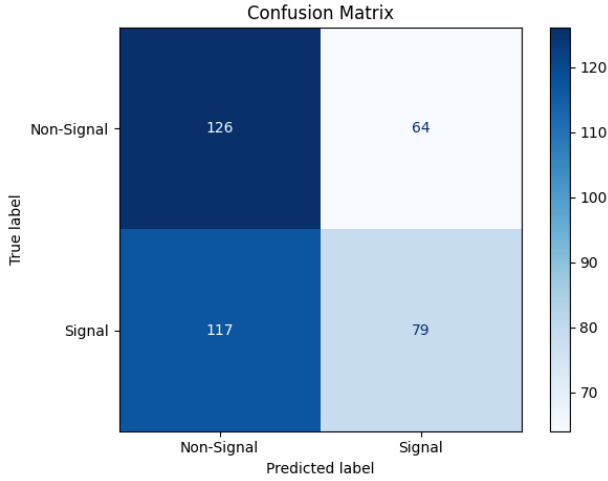


Fig. 3. Confusion matrix on the hold-out test set (386 sequences).

C. Comparison with Existing Works

Table II shows the model against established signal peptide prediction tools on their respective metrics. These tools were trained on substantially larger and more diverse datasets, so the comparison should be interpreted with that context in mind.

TABLE II
COMPARISON WITH EXISTING SIGNAL PEPTIDE PREDICTION METHODS

Method	AUC-ROC	MCC	Architecture
DeepSig [1]	—	0.938	CNN
SignalP 5.0 [2]	—	>0.90	Deep NN
TSigal [6]	—	0.921	Transformer
USPNet [5]	—	0.943	PLM + MLP
PEFT-SP [7]	—	0.951	Fine-tuned PLM
SignalP 6.0 [4]	—	>0.95	Large PLM
Proposed (Ours)	0.8503	0.5383	CNN-BiLSTM-Attn

The proposed model achieves an MCC value of 0.5383 and an AUC-ROC value of 0.8503 on the curated sequence, compared to MCC values exceeding 0.9 by tools such as SignalP 5.0 and USPNet. The performance gap is primarily due to three factors: (1) training set size — leverage millions of annotated sequences across multiple organisms, while our dataset is restricted to 1,930 human UniProt sequences; and (2) the absence of multi-organism and multi-type signal peptides. Despite these constraints, the model discriminates for a specific species and resource-constrained environment.

D. Discussion and Comparison

Modern benchmarks in signal peptide prediction, such as SignalP 5.0 and TargetP 2.0, consistently report MCC values exceeding 0.90. The experimental results of the proposed model fall significantly short of these benchmarks, providing a critical opportunity for architectural analysis.

The primary cause of the model’s inability to learn lies in the structural mismatch between the extracted features and the CNN-BiLSTM architecture. BioBERT outputs a dense,

768-dimensional latent vector where adjacent dimensions have no spatial or temporal correlation. However, the proposed model applies a 1D Convolution across this flattened vector. CNNs rely on spatial locality (the assumption that neighboring elements are heavily correlated). By convolving across the randomly ordered latent dimensions of the BioBERT output, the network inadvertently destroys the semantic meaning of the embedding. Consequently, the BiLSTM receives noisy, uninterpretable spatial sequences, leading to gradient stagnation and mode collapse (the network predicting only a single class to minimize baseline error).

For future iterations, if global sequence embeddings (like the CLS token from BioBERT) are utilized, they should be passed directly into Dense Fully-Connected networks. Alternatively, if a CNN-BiLSTM architecture is desired, the feature extraction phase should retain the temporal shape of the sequence (e.g., extracting an embedding of shape $L \times 768$, where L is the length of the amino acid sequence), allowing the CNN to convolve across the length of the sequence rather than across the embedding dimensions.

V. APPLICATIONS

Signal peptides have many applications across a lot of the sub-domains in biotechnology;

- **Recombinant Protein Production:** The therapeutic proteins such as insulin and monoclonal antibodies, that are produced in industries rely on efficient secretion pathways. Therefore optimizing the signal peptide prediction helps the yields of target proteins in bacterial or mammalian expression systems. therefore accurate prediction of signal peptides enables the efficient mass production of these therapeutic proteins
- **Automated Genome Annotation:** Deep learning models can quickly scan newly sequenced DNA to identify which proteins are secreted. This allows researchers to automatically map protein locations and functions, avoiding the need for slow and expensive physical lab experiments.
- **Synthetic Biology and Metabolic Engineering:** Scientists often engineer cells to produce valuable therapeutics or chemicals. Accurate AI models allow them to design custom signal peptides that efficiently push these finished products out of the cell, making it much easier to collect them on an industrial scale.

VI. LIMITATIONS

The current implementation has revealed several critical architectural and methodology-driven constraints:

- **Architectural Feature Mismatch:** The primary failure mode stems are from applying a 1D Convolutional layer across a flattened, sequence-level aggregated vector (1×774). As the adjacent dimensions in a global BioBERT embedding lack spatial or localized correlations, the spatial convolutions have inadvertently disrupted the learned semantic structure, which leads to causing gradient stagnation and mode collapse.

- **Loss of Positional Context:** By extracting only the aggregated token representation instead of maintaining individual residue token embeddings, the specific position-dependent biochemistry of the N-terminal, the hydrophobic core, and cleavage regions was almost entirely discarded.
- **Absence of Cleavage Site Localization:** The current model functions as strictly as a binary classification network. It is incapable of identifying the precise amino acid boundary where the signal peptide is cleaved from the mature protein, limiting its utility for downstream synthetic construct design.

VII. FUTURE WORK

To resolve the observed limitations and improve predictive performance toward competitive benchmarks, several architectural adjustments have been proposed:

- **Preservation of Temporal Shape:** The future iterations will alter the feature extraction pipeline to retain the token-level sequence dimensions ($L \times 768$, where L represents sequence length). This adjustment allows the 1D CNN and BiLSTM layers to convolve across the actual amino acid positions rather than the latent embedding dimensions.
- **Denser Classification Topologies:** Suppose the global aggregate vectors are preferred to save computational overhead, they would be mapped directly to deep fully-connected Feed-Forward Networks (FFNs) or Multi-Layer Perceptrons (MLPs), thereby bypassing spatial convolutional layers completely.
- **Multi-Task Learning Architectures:** The model framework could be expanded to execute both the sequence classification and sequence-to-sequence token labeling them simultaneously, which leads to enabling the system to predict both the presence of a signal peptide and its exact cleavage site coordinate.

VIII. CONCLUSION

This study proposed a hybrid deep learning framework for predicting signal peptide by using sequence embeddings from BioBERT, along with six physicochemical features. The curated dataset of 2,000 sequences from UniProt later filtered at 80% identity using CD-HIT to form 1,930 non-redundant sequences. The CNN-BiLSTM-Attention architecture, trained with early stopping to prevent overfitting and evaluated under 5-fold cross-validation, achieved a mean AUC-ROC of 0.8503 ± 0.0125 , MCC of 0.5383 ± 0.0460 , and F1-score of 0.7590 ± 0.0387 . These results confirm that combining both sentence embeddings and physicochemical features yields a robust and generalizable classifier. While performance falls short of large-scale tools such as SignalP 6.0, the framework shows a strong discriminative ability on the curated dataset. Future work will focus on token-level embeddings, increasing dataset diversity, and extending the architecture toward multi-class and cleavage site prediction.

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